

PROJECT REPORT

RETAIL SALES, CUSTOMERS & INVENTORY DASHBOARD USING MICROSOFT EXCEL

A Business Analytics Project Report

Bachelor of Business Administration (BBA)

Madras Christian College
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Submitted by
KARTHIKEYAN V

2401721096151

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CERTIFICATE

This is to certify that the project report entitled “Retail Sales, Customers & Inventory Dashboard Using Microsoft Excel” is a Bonafide academic project prepared as part of the Bachelor of Business Administration Programme. The report presents an analysis of retail sales, customer and inventory data using Microsoft Excel.

Date: _____

Place: Chennai

Project Guide: _____

Head of the Department: _____

DECLARATION

I hereby declare that the project report titled “Retail Sales, Customers & Inventory Dashboard Using Microsoft Excel” has been prepared for academic purposes based on the dataset used for this study. The analysis, interpretations and dashboard were prepared using Microsoft Excel.

Date: _____

Place: Chennai

Name: KARTHIKEYAN V

Signature: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to my faculty members and the Department of Business Administration for providing me with the opportunity to undertake this project. I am thankful for the guidance and encouragement received throughout the preparation of the project. I also thank my college for providing an academic environment that helped me understand the practical application of business analytics. Finally, I thank everyone who supported me in completing the analysis, dashboard and project documentation.

EXECUTIVE SUMMARY

This project analyses retail store sales, customer and inventory data using Microsoft Excel. The study uses 1,000 sales transactions, 300 customer records and 100 inventory product records. Excel formulas, PivotTables, Pivot Charts and dashboard visualization were used to convert raw data into useful business information.

Total sales were ₹1,42,946 from 1,000 transactions. Cleaning generated the highest category sales at ₹36,120, August recorded the highest monthly sales at ₹39,953, and Surf Excel was the top-selling product at ₹18,315. UPI represented approximately 51% of sales value.

There were 154 female and 146 male customers. Customers above 30 years formed the largest age group with 228 customers, while regular customers were the largest visit-type group with 104 customers. Cash was preferred by 175 customers.

Inventory analysis showed 13,507 units of stock and total inventory value of ₹9,65,948.25. Of 100 products, 92 had sufficient stock and 8 required reorder. Cleaning had the highest inventory value at ₹2,44,011.

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1. INTRODUCTION

Retail stores generate information through daily sales, customer purchases, payment transactions and inventory movements. Analyzing this information helps retailers understand product demand, sales trends, customer behavior and stock requirements. This project applies to Microsoft Excel to develop a practical dashboard combining sales, customer and inventory analysis.

Need for the Study

- Convert raw retail records into useful information.
- Identify sales trends and high-performing products.
- Understand customer segments and payment preferences.
- Monitor inventory availability and inventory investment.
- Present important results through a visual dashboard.

2. BUSINESS PROFILE

The project represents a retail/provisional store environment selling frequently purchased consumer products. The major categories are Beverages, Biscuits, Grocery, Cleaning and Personal Care. Typical activities include sales, billing, customer service, payment collection, stock monitoring and product replenishment.

Activity	Purpose
Sales and Billing	Record customer purchases and sales value.
Customer Service	Serve customers and support repeat purchases.
Inventory Management	Monitor stock and reorder requirements.
Payment Collection	Handle Cash and UPI transactions.
Product Monitoring	Track product and category performance.

3. PROBLEM STATEMENT

Retail stores need a simple system to monitor sales, customers and inventory simultaneously. Without systematic analysis, it can be difficult to identify high-performing products, understand customer groups, detect sales trends and recognize products requiring replenishment. This project addresses the problem through an Excel-based dashboard.

4. OBJECTIVES

1. Measure overall retail sales performance.
2. Analyze monthly and category-wise sales.
3. Identify top-selling products.
4. Understand customer gender, age group and visit type.
5. Analyze payment preferences.
6. Evaluate inventory stock status and inventory value.
7. Identify products requiring reorder.
8. Develop an Excel dashboard for management reporting.

5. SCOPE

The study covers sales performance, product categories, monthly sales, top products, payment modes, customer gender, age groups, visit type, inventory stock status and inventory value. The project is limited to the fields available in the prepared datasets and is intended for academic business analytics.

6. METHODOLOGY

6.1 Research Design

A descriptive analytical approach was used to summarize and interpret existing retail sales, customer and inventory data.

6.2 Data Used

Dataset	Records	Purpose
Sales Data	1,000	Sales and product performance
Customer Data	300	Customer characteristics and preferences
Inventory Data	100	Stock and inventory value

6.3 Tools Used

- Microsoft Excel
- Excel formulas
- PivotTables
- Pivot Charts
- Sorting and filtering
- Dashboard visualization

6.4 Analytical Process

1. Organize datasets.
2. Check fields and data structure.
3. Calculate required metrics.
4. Create PivotTables.
5. Create Pivot Charts.
6. Arrange dashboard.
7. Interpret results and prepare recommendations.

7. DATA DESCRIPTION

7.1 Sales Data

Contains Date, Bill No, Customer ID, Product ID, Product Name, Category, Quantity, Unit Price, Discount, Sales Amount and Payment Mode.

7.2 Customer Data

Contains Customer ID, Customer Name, Gender, Age Group, Age, Visit Type and Preferred Payment.

7.3 Inventory Data

Contains Product ID, Product Name, Category, Brand, Purchase Price, Selling Price, Stock Available, Reorder Level and Expiry Date.

8. DATA PREPARATION AND EXCEL TECHNIQUES

The datasets were structured in Excel and summarized using PivotTables. Pivot Charts were used to present results visually. The dashboard uses KPI cards and charts for quick management review.

Metric	Formula / Method
Total Sales	SUM of Sales Amount
Transactions	COUNT of transaction records
Customers	COUNT of Customer IDs
Total Stock	SUM of Stock Available
Inventory Value	SUM of Purchase Price $\times$ Stock Available
Stock Status	Compare Stock Available with Reorder Level
Average Transaction Value	Total Sales $\div$ Total Transactions

9. SALES ANALYSIS

Analysis	Result	Interpretation
Total Sales	₹1,42,946	Overall sales value.
Transactions	1,000	Number of recorded transactions.
Category-wise Sales	Cleaning – ₹36,120	Highest category sales.
Monthly Sales	August – ₹39,953	Highest monthly sales.
Top Product	Surf Excel – ₹18,315	Highest-selling product.
Payment Mode	UPI – ₹73,334 (51%)	Slightly higher than Cash in sales value.
Quantity by Category	Beverage – 823 units	Highest quantity sold.

Interpretation: Cleaning is an important revenue contributor. August was the strongest month in the dataset, while Surf Excel was the leading product by sales value. Both Cash and UPI are important payment methods.

10. CUSTOMER ANALYSIS

Analysis	Result	Interpretation
Gender	Female 154; Male 146	Female customers were slightly higher.
Age Group	Above 30 – 228	Largest age segment.
Visit Type	Regular – 104	Largest visit-type group.
Preferred Payment	Cash 175; UPI 125	Cash was preferred by more customers.

Interpretation: The customer base is fairly balanced by gender. The above-30 segment is the largest group, and regular customers form the largest visit segment, highlighting the importance of customer retention.

11. INVENTORY ANALYSIS

Analysis	Result	Interpretation
Total Stock	13,507 units	Total available inventory.
Inventory Value	₹9,65,948.25	Total inventory investment.
Stock Status	92 sufficient; 8 reorders required	8 products require replenishment.
Highest Inventory Value	Cleaning – ₹2,44,011	Highest category inventory investment.

Interpretation: Most products have sufficient stock, but 8 products require replenishment. High-value inventory, particularly Cleaning, should be monitored to avoid excess working capital being tied up in stock.

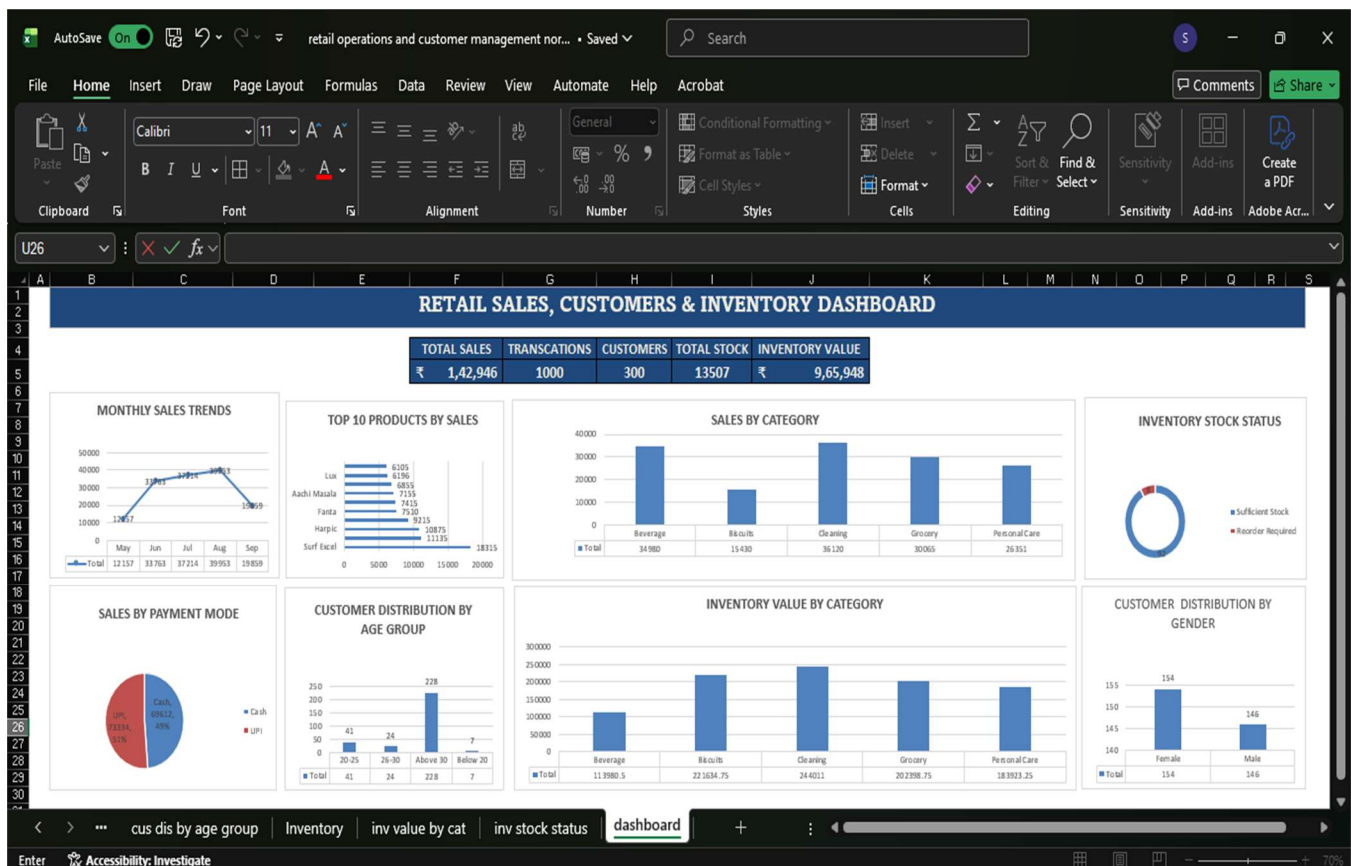
12. DASHBOARD DEVELOPMENT

The final dashboard combines sales, customer and inventory results. Five KPI cards are placed at the top, followed by charts covering monthly sales trends, top products, category sales, payment mode, customer age group, customer gender, inventory stock status and inventory value by category.

Dashboard Component	Purpose
Total Sales	Overall sales value.
Transactions	Number of transactions.
Customers	Customer base size.
Total Stock	Total units available.
Inventory Value	Total inventory investment.

Monthly Sales Trends	Monthly sales movement.
Top 10 Products by Sales	Leading products.
Sales by Category	Category comparison.
Sales by Payment Mode	Cash vs UPI sales.
Customer Distribution by Age Group	Age-segment comparison.
Customer Distribution by Gender	Gender comparison.
Inventory Stock Status	Sufficient vs reorder-required stock.
Inventory Value by Category	Inventory investment comparison.

DASHBOARD SCREENSHOT



13. KEY FINDINGS

- Total sales amounted to ₹1,42,946 of 1,000 transactions.
- Cleaning generated the highest category sales of ₹36,120.
- August recorded the highest monthly sales of ₹39,953.
- Surf Excel was the top-selling product with sales of ₹18,315.
- UPI accounted for approximately 51% of sales value.
- Female customers numbered 154 compared with 146 male customers.
- Customers above 30 years formed the largest age group with 228 customers.
- Regular customers were the largest visit-type group with 104 customers.
- Cash was preferred by 175 customers, compared with 125 preferring UPI.
- Total inventory stock was 13,507 units.
- 8 products required reorders while 92 had sufficient stock.
- Total inventory value was ₹9,65,948.25.
- Cleaning had the highest inventory value at ₹2,44,011.

14. RECOMMENDATIONS

1. Maintain sufficient stock of high-selling products such as Surf Excel.
2. Recorder the 8 products identified as requiring replenishment.
3. Monitor the Cleaning category because it has high inventory value.
4. Study the reasons behind strong August sales and apply successful practices for other months.
5. Continue supporting both cash and UPI payment methods.
6. Develop retention offers for regular customers.
7. Use targeted promotions for the large above-30 customer segment.
8. Review lower-performing categories and consider suitable promotions.
9. Monitor inventory value regularly to prevent excessive capital from being tied up in stock.

15. CONCLUSION

The project successfully demonstrated how Microsoft Excel can be used to analyze retail sales, customers and inventory in one integrated business analytics project. The analysis converted raw records into KPIs, PivotTables, charts and a management dashboard. The results identify important sales trends, customer segments and inventory requirements. Overall, the dashboard provides a simple and practical approach to support retail decision-making.

16. LIMITATIONS

- The study uses a prepared data set and may not represent every real-world retail situation.
- The analysis is limited to the available fields.
- External factors such as competitors and market conditions were not included.
- Results depend on the accuracy of input data.
- The project focuses mainly on descriptive analysis.

17. FUTURE SCOPE

- Develop the dashboard further in Microsoft Power BI.
- Integrate real-time sales and inventory data.
- Introducing sales forecasting.
- Expand customer segmentation.
- Automate recommendations.
- Add profit margin and supplier performance analysis.

18. REFERENCES

- Microsoft Excel Support and Documentation.
- Business Analytics textbooks and classroom materials.
- Retail Management textbooks and academic materials.
- Research Methodology textbooks and academic materials.

APPENDIX

Project title: Retail Sales, Customers & Inventory Dashboard Using Microsoft Excel

Primary tool: Microsoft Excel

Dataset size: 1,000 sales transactions, 300 customers and 100 inventory products.

Final deliverables: Excel workbook, dashboard screenshot and project report.

links

[retail operations and customer management normal .xlsx](#)